DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) — ROUND 5

SentiLingo

Multilingual Customer Sentiment Analysis Platform



Project Documentation

Part 1 Project Planning & Management
Part 2 Literature Review
Part 3 Requirements Gathering
Part 4 Database Design
Part 5 UI/UX Design

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Version	1.3 — 3 November 2026

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PART 1

1. Project Planning & Management

What SentiLingo is, what it will deliver and when, how the team works, which risks matter most and how success is measured.

1.1 Project Overview

Companies receive customer feedback through app and e-commerce reviews, social media, surveys and support tickets. In Egypt this feedback is commonly written in Modern Standard Arabic (MSA), Egyptian Arabic and English, and may mix Arabic and English in one message. Tagging it by hand is slow and inconsistent, and many ready-made tools do not report how well they handle Egyptian Arabic (see Part 2).

> **Problem statement (from the project brief).** A company receives customer feedback in natural-language text and needs to automatically identify whether feedback expresses positive, negative, or other predefined sentiment categories.

SentiLingo labels each piece of feedback **Negative**, **Neutral** or **Positive** with an estimated confidence. It compares a dummy classifier and focused classical baselines using word, character and combined TF-IDF with Multinomial Naive Bayes, Logistic Regression and Linear SVM. XLM-R Base is the required advanced model; MARBERT is an Arabic-only stretch experiment and BiLSTM is included only if the instructor requires it. Models are selected using the mean of the English, MSA and Egyptian Arabic macro-F1 scores on validation data. The locked configuration is then evaluated once on the frozen test set.

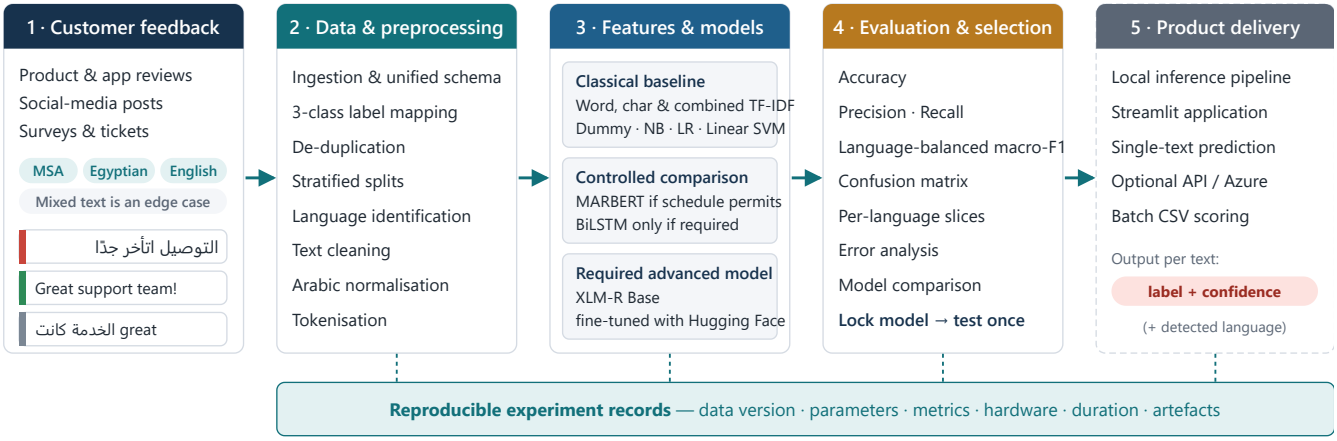


Figure 1 — SentiLingo solution overview.

1.2 Objectives

ID	Objective	Due (2026)
O1	Approve project planning, stakeholder, database and UI/UX design artefacts; select licensed primary and backup datasets.	8 Nov
O2	Build a unified English, MSA and Egyptian Arabic dataset with three sentiment classes, frozen splits and a data card.	15 Nov
O3	Compare word, character and combined TF-IDF using a dummy classifier, Multinomial Naive Bayes, Logistic Regression and Linear SVM.	15 Nov
O4	Fine-tune XLM-R Base; attempt MARBERT or BiLSTM only under the documented stop rule.	22 Nov
O5	Select the model on validation data, evaluate the locked configuration once on the test set and complete error analysis.	29 Nov
O6	Deliver a reliable local Streamlit application for single-text and CSV prediction.	29 Nov
O7	Deliver the repository, documentation, model card, presentation PDF, video demonstration and offline backup.	4 Dec

1.3 Scope

In scope

- English, Modern Standard Arabic and Egyptian Arabic
- Three classes: Negative, Neutral, Positive
- Public, labelled review and tweet data
- Word, character and combined TF-IDF with dummy, NB, LR and Linear SVM baselines
- XLM-R Base, reproducible experiment records, evaluation and error analysis
- Local Streamlit application with single-text and CSV prediction

Out of scope

- Aspect-based sentiment and emotion detection
- Speech, image or video feedback
- Live social-media crawling and CRM integration
- Collecting private customer data
- Production SLAs and automated retraining
- Franco-Arabic and Arabic dialects other than Egyptian Arabic
- User accounts, authentication and a production database
- FastAPI, Docker and Azure unless the core release is complete

Assumptions and constraints

- The five official team members have limited availability, so the plan protects a small required release and uses one owner and one reviewer for each deliverable.
- Zero budget: local resources, free Colab or Kaggle GPUs, and open-source tools; Azure remains optional.
- Only public data is used, and no personal data is committed to the repository.
- The local RTX 4050 with 6 GB VRAM is used for smoke tests and inference; hosted GPUs are used for the main Transformer runs.
- The active five-week delivery period starts on 2 November and the final presentation is on 4 December 2026.

1.4 Deliverables

ID	Deliverable (from the brief)	Description	Milestone
D1	Planning and design package	Project planning, stakeholder analysis, database design and UI/UX design	MS0 · 2 Nov
D2	Dataset and data card	Source register, licences, label mapping, common schema, quality audit and frozen splits	MS2 · 15 Nov
D3	NLP preprocessing	Reproducible cleaning, Arabic normalisation and feature preparation with leakage checks	MS2 · 15 Nov
D4	Classical baselines	Dummy, NB, LR and Linear SVM with word, character and combined TF-IDF	MS2 · 15 Nov
D5	Advanced model	XLM-R Base plus an optional controlled MARBERT or BiLSTM comparison	MS3 · 22 Nov
D6	Evaluation and error analysis	Validation leaderboard, one locked test evaluation, confusion matrices and error taxonomy	MS4 · 29 Nov
D7	Local application	Streamlit single-text and CSV prediction with Arabic right-to-left display	MS4 · 29 Nov
D8	Reproducibility package	Experiment records, tests, environment instructions, data card and model card	MS5 · 4 Dec
D9	Final documentation	README and consolidated technical documentation with decisions and limitations	MS5 · 4 Dec
D10	Presentation package	Presentation PDF, video demonstration, rehearsal and offline backup	MS5 · 4 Dec

1.5 Methodology

The project follows CRISP-DM (Wirth & Hipp, 2000): business understanding, data understanding, data preparation, modelling, evaluation and deployment. The work is compressed into five weekly delivery cycles. Each cycle begins with a short planning meeting, assigns one owner and one reviewer to every deliverable, and ends with evidence-

based review. GitHub issues track work, reviewed pull requests protect the main branch, and every reported result records its data version, configuration, seed, hardware, duration, metrics and artefact location.

1.6 Timeline & Milestones

The active delivery plan runs from **2 November** to the final presentation on **4 December 2026**, in five focused weeks (W1–W5). The final date is fixed; internal milestones are stop points that prevent optional work from displacing the required release.

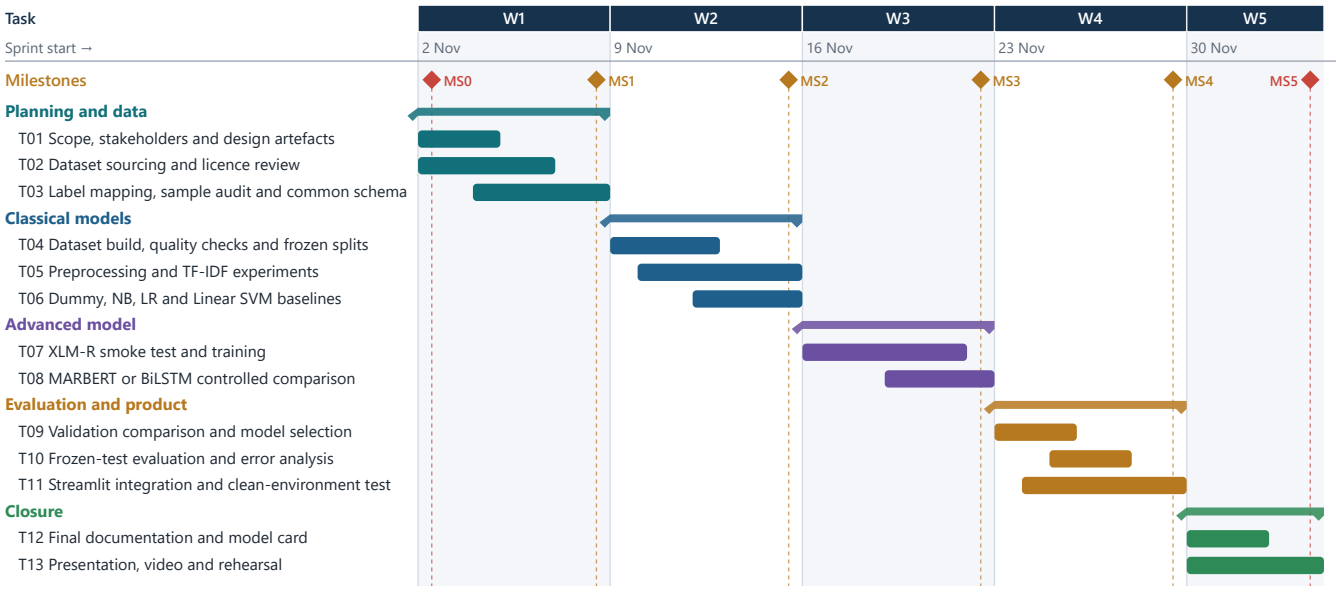


Figure 2 — Project schedule. Bars are coloured by phase; ◆ marks a milestone (red = fixed date).

ID	Milestone	Target date	Exit criteria (evidence)	Type
MS0	**Planning package and repository aligned**	2 Nov 2026	Project plan, stakeholders, database design, UI/UX design and repository links are prepared	Official DEPI
MS1	**Data decisions approved**	8 Nov 2026	Primary and backup sources, licences, label mapping, common schema and sample audit are approved	Internal
MS2	**Dataset and classical baselines ready**	15 Nov 2026	Frozen splits, data card and comparable dummy, NB, LR and Linear SVM results exist	Internal
MS3	**Advanced model ready**	22 Nov 2026	XLM-R is trained and evaluated on validation data; optional comparison is either completed or formally dropped	Internal
MS4	**Evaluation and local demo ready**	29 Nov 2026	Model is locked, test evaluation is complete, errors are analysed and the Streamlit demo works locally	Internal
MS5	**Final presentation**	4 Dec 2026	Documentation, repository, PDF presentation, video demonstration and offline backup are delivered	Official DEPI

1.7 Team & Task Distribution

The team has five members:

1. احمد مؤنس احمد عبدالفتاح
2. نوران ناصر جلال سيد
3. محمد محمود محمد عبدالله عياد
4. احمد اشرف سامى فريد
5. رضوه أشرف جمعه مهدي

The official team has five members. Work is split into five accountable areas, but ownership may rotate as availability changes. Every deliverable has one owner, one reviewer and a visible completion criterion. No task is considered complete only because a file exists.

Work area	Main tasks
Product and planning	Scope, stakeholders, backlog, risks, design decisions and delivery evidence
Data and quality	Dataset sourcing, licence checks, label mapping, sample audit, schema, splits and data card
Classical modelling	Preprocessing, word and character TF-IDF, dummy, NB, LR and Linear SVM experiments
Advanced modelling and evaluation	XLM-R training, optional comparison, experiment records, model selection and error analysis
Application and communication	Streamlit, tests, README, technical report, presentation, video and rehearsal

Communication: one weekly planning and review meeting, brief asynchronous status updates, GitHub issues for ownership and acceptance criteria, and pull requests for review. Optional tasks require explicit approval after required work is on schedule.

1.8 Risk Management

Risks are reviewed at every sprint review. The level combines likelihood and impact before mitigation.

ID	Risk	Level	Mitigation
R1	Too little labelled Egyptian-dialect customer feedback	High	Combine several public Arabic sources; label a small gold set by hand; report the dialect slice separately
R2	A dataset is withdrawn or has a restrictive licence	High	Check licences in week 1; keep backup sources; commit download scripts, not raw data
R3	Limited GPU memory, hosted-GPU quota or interrupted sessions	High	Use local CPU for classical work, local GPU for smoke tests and inference, hosted GPU for main runs, small batches and saved checkpoints
R4	Data leakage: duplicates across splits, or preprocessing fitted on all data	High	De-duplicate before splitting; scikit-learn Pipelines fitted on training data only; frozen test set
R5	Inconsistent labels across sources, and too few Neutral examples	Medium	Written label-mapping policy; manual audit; stratified splits; class weights; macro-F1 as the main metric
R6	Egyptian spelling variation, mixed text, sarcasm and negation cause errors	Medium	Arabic normalisation, character n-grams, language-specific evaluation, difficult-example audit and explicit limitations
R7	Uneven availability of team members and delayed reviews	High	One owner and one reviewer per deliverable, small tasks, early escalation and required-work stop rules
R8	Optional API, Docker, Azure or extra models consume the final week	Medium	Keep Streamlit local-first and start optional work only after evaluation, documentation and rehearsal are safe

1.9 Key Performance Indicators

ID	Key performance indicator	Success rule
K1	Primary validation score	**Mean of English, MSA and Egyptian Arabic macro-F1**; no language may be hidden by the largest subset
K2	Evidence over a trivial baseline	The selected model must beat the dummy classifier and be compared fairly with the strongest classical baseline
K3	Per-language and per-class transparency	Macro-F1, class metrics and confusion matrices are reported for all three supported varieties
K4	Prediction usability	Single-text and CSV prediction complete reliably on the presentation laptop; latency and model size are recorded
K5	Experiment traceability	**100%** of reported results identify data, configuration, seed, hardware, duration and artefacts
K6	Required milestone protection	Required work is completed before any optional model or cloud deployment begins
K7	Delivery completeness	Repository, documentation, presentation PDF, video and offline demo backup all open from the submitted links

Validation data selects the model; the frozen test set is evaluated once after all preprocessing, model and threshold decisions are locked.

PART 2

2. Literature Review

Related research and existing solutions, what SentiLingo borrows from them, and where it differs.

2.1 Review Method

We searched the ACL Anthology, Google Scholar, arXiv and vendor documentation for work on the sentiment of reviews and social-media text, focusing on Arabic, Egyptian dialect and multilingual models. Every figure quoted below was checked against the original source (October 2026).

2.2 Background and Challenges

Sentiment analysis classifies the opinion expressed in a text. Customer feedback makes this hard: messages are short and informal, often mix praise and complaints, and neutral messages are hard to label consistently. Arabic adds further difficulties:

- rich morphology and inconsistent spelling (e.g. ا/أ, ي/ى);
- dialects such as Egyptian Arabic that differ from MSA in vocabulary and grammar;
- Franco-Arabic and code-switching with English, for which very few labelled resources exist (Hamed et al., 2025);
- sarcasm and label noise, the main sources of the errors that remain in Arabic sentiment models (Abu Farha & Magdy, 2021).

2.3 Classical Approaches

Lexicon-based tools such as VADER score text with a hand-built word list plus rules for negation and intensity. They need no training data but are English-only (Hutto & Gilbert, 2014).

Supervised learning does better:

- On movie reviews, Naive Bayes and SVM classifiers on word features reached 81–83% accuracy and beat keyword lists (Pang et al., 2002).
- Adding word bigrams lifted a Naive Bayes–SVM hybrid to 91.2% (Wang & Manning, 2012).
- For Arabic, an SVM on TF-IDF n-grams scored 0.63 F1 on the four-class Egyptian tweet corpus ASTD (Nabil et al., 2015).
- The best Arabic system at SemEval-2017 was a Naive Bayes model (Rosenthal et al., 2017).

These models are fast, cheap, interpretable and effective with modest data. However, they see no context beyond n-grams, are weak on negation and sarcasm, and break on unseen dialect spellings. This is the baseline required by the brief; SentiLingo adds character n-grams to handle spelling variation.

2.4 Deep Learning and Transformers

LSTMs (Hochreiter & Schmidhuber, 1997) read text as a sequence and can learn effects such as the scope of a negation. They need more data and tuning than linear models.

Pretrained Transformers are first trained on large text corpora and then fine-tuned for the task:

- BERT reached 94.9% on a standard English sentiment benchmark (Devlin et al., 2019).
- XLM-R extends this approach to 100 languages (Conneau et al., 2020).
- XLM-T, a Twitter-adapted version of XLM-R, reached macro-F1 66.9 on Arabic and 70.6 on English three-class tweets (Barbieri et al., 2022).

Arabic-specific models do better on Arabic:

- AraBERT reached 92.6% accuracy on the balanced ASTD, against 80.1% for multilingual BERT (Antoun et al., 2020).

- MARBERT, pretrained on one billion dialect-rich tweets, beat both XLM-R and multilingual BERT on SemEval-2017 Arabic (Abdul-Mageed et al., 2021).

Large language models can classify sentiment without any training. They do well on simple tasks but trail fine-tuned smaller models on harder ones, and they cost much more per prediction (Zhang et al., 2024).

2.5 Arabic and Egyptian-Dialect Studies

The study closest to SentiLingo compared a TF-IDF SVM, an LSTM, multilingual BERT and Arabic BERT models on 40,000 Egyptian tweets (Abu Kwaik et al., 2022). The SVM (0.84 accuracy) matched multilingual BERT (0.83), a Twitter-adapted Arabic BERT was best (0.91) and the LSTM was weakest (0.49); the study used two classes only. A re-implementation of published Arabic methods found that Transformers gave the best F-scores: 0.69 on SemEval-2017 and 0.76 on ASTD (Abu Farha & Magdy, 2021). For code-switched feedback, a corpus of 4,100 Egyptian Arabic–English comments reached an F1 of about 0.93, both with neural models and with a fine-tuned GPT-3.5 (Sherif & Sabty, 2025).

2.6 Existing Solutions

Solution	Type	Arabic support	Strengths	Weaknesses
Azure Language sentiment analysis (Microsoft, 2026)	Cloud API	Generic Arabic	Managed; labels with confidence per sentence and per document	Cannot be customised to the user's data; the feature retires on 31 Mar 2029
Amazon Comprehend (Amazon Web Services, n.d.)	Cloud API	Generic Arabic	Managed; batch jobs; four labels including Mixed	Entity-level sentiment in English only; pay per use
Google Cloud Natural Language (Google Cloud, n.d.)	Cloud API	Generic Arabic	Simple score and magnitude for each sentence	No class labels, so thresholds must be set; no customisation within the API
VADER (Hutto & Gilbert, 2014)	Open-source lexicon	None	Free, fast, no training needed	English only; misses domain and dialect words
Twitter-XLM-R sentiment model (Barbieri et al., 2022)	Open pretrained model	8 languages, including Arabic	One multilingual model, ready to use	Trained on tweets; about 0.67 macro-F1 on Arabic

Table 2.1 — Ready-made options. None of the cloud APIs lists Egyptian Arabic or Franco-Arabic as a supported variety.

2.7 Gap Analysis

#	Gap in existing work	SentiLingo response
G1	Egyptian dialect and Franco-Arabic are poorly covered by ready-made tools	The core pipeline supports MSA, Egyptian Arabic and English with separate evaluation; Franco-Arabic is documented as out of scope
G2	Classical and Transformer models are rarely compared on the same data; studies often use one dataset and two classes	The same frozen splits, seeds and metrics for every model family, with three classes
G3	Commercial APIs cannot be customised and can be retired	An owned model that can be retrained on company data
G4	Published results are hard to reproduce	Every reported run has a structured experiment record, with reproduction steps in the README; MLflow may implement the record
G5	LLMs are costly per prediction	Fine-tuned compact models, with latency reported next to accuracy

2.8 Lessons Applied

1. Tune the classical baseline carefully, with word and character n-grams, and keep negation words.
2. Compare a multilingual Transformer with a dialect-aware Arabic model.
3. Report macro-F1 and per-class metrics, with realistic targets: about 0.67–0.71 macro-F1 is already strong for three-class tweets.

PART 3

3. Requirements Gathering

Who SentiLingo serves, what it must do, how well it must do it, and the data and tools it needs.

3.1 Elicitation Approach

Requirements were consolidated from five evidence sources:

1. the DEPI submission request and course learning outcomes;
2. the approved scope and design decisions in Part 1;
3. the research and existing-solution review in Part 2;
4. inspection of candidate datasets, licences and sample records;
5. delivery constraints: five weeks, limited team availability, local-first demonstration and zero budget.

Instructor confirmation is still required for the exact submission deadline and whether Azure or BiLSTM is mandatory. Until that evidence exists, neither is allowed to displace the required local release. Each requirement has an ID and a MoSCoW priority: Must, Should or Could.

3.2 Stakeholder Analysis

The primary business user is the **customer-experience manager**, who needs a practical way to review multilingual feedback. The DEPI instructor and evaluators have the greatest formal influence because they judge scope, evidence and delivery. The project team owns implementation decisions, while dataset providers constrain lawful use.

Stakeholder	Influence	Interest	Main need	Engagement and evidence
DEPI instructor and evaluators	High	High	Rubric compliance, defensible methodology, honest results and complete links delivered on time	Confirm open requirements; provide documentation, repository evidence, presentation and demonstration
Five-person project team	High	High	Clear ownership, achievable workload, shared learning and early warning when work slips	Weekly planning and review; GitHub issues, pull requests and decision log
Customer-experience manager	Medium	High	Understand overall and per-language sentiment, analyse CSV feedback and export useful results	Validate the primary workflow, output columns, explanations and representative examples
Support employee or demo user	Low	High	Paste one English, MSA or Egyptian Arabic message and receive a clear result without technical knowledge	Review wireframes and test empty, Arabic, mixed-text and low-confidence cases
Dataset and model providers	Medium	Medium	Licence compliance, attribution and no unauthorised redistribution	Maintain source register, licence record, citations and raw-data exclusion policy
Future technical reviewers	Low	Medium	Readable repository, reproducible experiments and honest limitations	README, data card, model card, experiment evidence and clean-environment test

Power and Interest Strategy

Strategy	Stakeholders	Project response
Manage closely High influence, high interest	DEPI instructor and evaluators; project team	Resolve requirements early, expose risks, review evidence weekly and obtain explicit approval for scope changes
Keep satisfied High influence, lower daily interest	Programme administration; dataset and model providers	Submit accessible links on time and comply with licences, attribution and redistribution restrictions
Keep informed Lower influence, high interest	CX manager; support employee or demo user	Validate the user flow, labels, CSV contract, Arabic display, explanations and error behaviour

Strategy	Stakeholders	Project response
Monitor Lower influence, lower immediate interest	Future technical reviewers	Keep documentation, repository history and reproducibility evidence understandable without adding portfolio-only scope

Team Responsibility Model

Each official member is assigned one operating role in the weekly plan. Names may rotate between roles, but every work item must retain exactly one accountable owner. R = responsible, A = accountable, C = consulted and I = informed.

Work item	Project coordinator	Data owner	Model owner	Application owner	Documentation owner
Scope, schedule, risks and stakeholder decisions	**A/R**	C	C	C	R
Dataset sources, licences, schema, audit and splits	A	**R**	C	I	C
Preprocessing, classical and advanced experiments	I	C	**A/R**	C	I
Validation policy, locked test and error analysis	A	C	**R**	I	C
Prediction contract, Streamlit and integration tests	C	I	C	**A/R**	C
README, report, presentation, video and link verification	A	I	C	C	**R**

3.3 User Stories

ID	User story	Acceptance criteria
US-01	As a **CX manager**, I want to upload a CSV of feedback and receive a sentiment result for every valid row.	The interface validates the required text column and exports row ID, language variety, label, confidence estimate and any error message.
US-02	As a **support employee**, I want to paste one English, MSA or Egyptian Arabic message and understand the prediction.	Arabic is displayed right-to-left; the result shows label, confidence estimate, detected variety and a limitation notice.
US-03	As an **ML engineer**, I want each experiment recorded so that I can compare or reproduce it.	The record identifies data and split versions, preprocessing, model, parameters, seed, hardware, duration, metrics and artefact location.
US-04	As a **project evaluator**, I want models compared fairly without test-set tuning.	The validation leaderboard uses language-balanced macro-F1; the final report records one frozen-test evaluation after locking the configuration.
US-05	As a **presentation reviewer**, I want the local demonstration to work without internet access.	A clean-environment checklist and offline fallback reproduce representative single-text and CSV predictions.

3.4 Functional Requirements

ID	Requirement	Priority
FR-01	Load labelled data from several sources into one schema: text, label, language, source, split.	Must
FR-02	Map each source's labels or star ratings to Negative, Neutral and Positive using a documented policy.	Must
FR-03	Remove duplicates, then create stratified train / validation / test splits with a fixed seed. The test set is frozen.	Must
FR-04	Preserve or assign one supported variety to each record: English, MSA or Egyptian Arabic; flag mixed or uncertain text for analysis.	Must
FR-05	Clean text: mask URLs, mentions, e-mails and phone numbers; strip HTML; handle emojis; lower-case Latin script.	Must

ID	Requirement	Priority
FR-06	Normalise Arabic: remove diacritics and tatweel; unify alef forms (آ إ أ → ا) and ى → ي; reduce repeated letters.	Must
FR-07	Prepare word and character inputs for classical models and subword-tokenised inputs for Transformers while preserving negation evidence.	Must
FR-08	Compare word TF-IDF, character TF-IDF and their combined representation.	Must
FR-09	Train a dummy classifier, Multinomial Naive Bayes, Logistic Regression and Linear SVM on the same frozen splits.	Must
FR-10	Train a BiLSTM only when the instructor confirms it is mandatory or the core schedule remains safe.	Could
FR-11	Fine-tune XLM-R Base as the required advanced model; attempt MARBERT on the Arabic subset only under the documented stop rule.	Must
FR-12	Report the mean of per-language macro-F1 as the primary validation score, with accuracy, overall macro-F1, weighted F1, per-class metrics and confusion matrices as supporting evidence.	Must
FR-13	Select the full configuration using validation data, then evaluate the locked configuration once on the frozen test set.	Must
FR-14	Record every reported experiment with data version, split version, preprocessing, model, parameters, seed, hardware, duration, metrics and artefact location; MLflow may implement this record.	Must
FR-15	Produce an error analysis and a table of sample predictions.	Must
FR-16	Predict single texts and CSV files, returning a label, confidence estimate, supported variety and validation message.	Must
FR-17	Provide a local Streamlit application with Arabic right-to-left display, input validation, batch export and an offline presentation path.	Must
FR-18	Add FastAPI, Docker or Azure deployment only after the local application, evaluation, documentation and rehearsal are stable.	Could
FR-19	Provide a README and a model card.	Must

3.5 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Evaluation	Model selection uses the approved validation metric and never uses final test results for tuning.
NFR-02	Performance	Single-text and CSV latency, memory use and model size are measured on the presentation environment before a performance target is claimed.
NFR-03	Security	No API keys or tokens in the repository; secrets come from environment variables.
NFR-04	Privacy	Personal data (handles, e-mails, phone numbers) is masked, and raw third-party data is never committed.
NFR-05	Usability	A first-time user can complete single-text and CSV prediction from labelled controls; Arabic is displayed right-to-left and sentiment is not communicated by colour alone.
NFR-06	Reproducibility	Fixed seeds, pinned dependencies and versioned splits; a re-run reproduces macro-F1 within ±0.01.
NFR-07	Reliability	Empty, emoji-only, mixed-script or very long inputs never crash the pipeline.
NFR-08	Maintainability	Modular code in `src/`, with unit tests for preprocessing and evaluation.
NFR-09	Transparency	Results are reported separately for English, MSA and Egyptian Arabic, and material performance gaps are explained rather than hidden by an overall score.
NFR-10	Compliance	Datasets and models are used under their licences, and all sources are cited.

3.6 Data Requirements

Label	Definition	English example	Arabic example
Negative	Dissatisfaction, a complaint or criticism	"The order arrived broken."	الطلب وصل مكسور
Neutral	No clear sentiment: facts, questions or balanced opinions	"When does the store open?"	المحل بيفتح الساعة كام؟
Positive	Satisfaction, praise or a recommendation	"Fast delivery, thank you!"	التوصيل سريع، شكرًا

Label mapping: star ratings 1–2 map to Negative, 3 to Neutral and 4–5 to Positive unless source documentation justifies a different policy. Ambiguous, mixed-sentiment and conflicting examples are flagged for review rather than silently forced into Neutral.

Candidate dataset	Language	Domain	Size	Licence	Planned use
HARD (Elnagar et al., 2018)	MSA and dialects	Hotel reviews	≈ 374k reviews, rated 1–5	None declared	Main Arabic training data
ASTD (Nabil et al., 2015)	Egyptian and MSA	Tweets	10,006	GPL-2.0	Egyptian-dialect training data
Twitter US Airline Sentiment (CrowdFlower, n.d.)	English	Customer-service tweets	14,640	CC BY-NC-SA 4.0	Main English training data
SemEval-2017 Task 4A (Rosenthal et al., 2017)	Arabic, English	Tweets	9,455 Arabic; 62,617 English	Free use beyond SemEval	Extra three-class training data and a benchmark
UMSAB (Barbieri et al., 2022)	8 languages, including Arabic and English	Tweets	3,030 per language	CC BY 3.0	External multilingual benchmark
JERD — Jumia Egypt Reviews (aliallam13, 2026)	Egyptian, MSA, English, Franco-Arabic	E-commerce reviews	3,809	CC BY-NC-ND 4.0	Candidate real-feedback source; use only supported-language records and respect no-derivatives restrictions

- Every supported-language evaluation slice must contain all three labels; minimum sample targets are set only after dataset inspection.
- Train, validation and test splits are stratified by language and label where counts allow. The test set is frozen before model experiments.
- Duplicates are removed before splitting, licences are recorded for each source, and raw data is not committed.

3.7 Technology Stack

Layer	Tools
Language and environment	Python 3.10+, Jupyter, Google Colab, Kaggle
Data and preprocessing	pandas, Hugging Face Datasets, regular expressions, NLTK, Arabic normalisation utilities
Classical ML	scikit-learn: TF-IDF, DummyClassifier, Multinomial Naive Bayes, Logistic Regression, Linear SVM and Pipelines
Advanced NLP	Hugging Face Transformers on PyTorch: XLM-R Base; optional MARBERT and conditional BiLSTM comparison
Experiment tracking	Structured experiment records; MLflow may be used when it does not delay required work
Application and collaboration	Local Streamlit; GitHub issues and pull requests; optional FastAPI, Docker and Azure after the core release

3.8 Use-Case Model

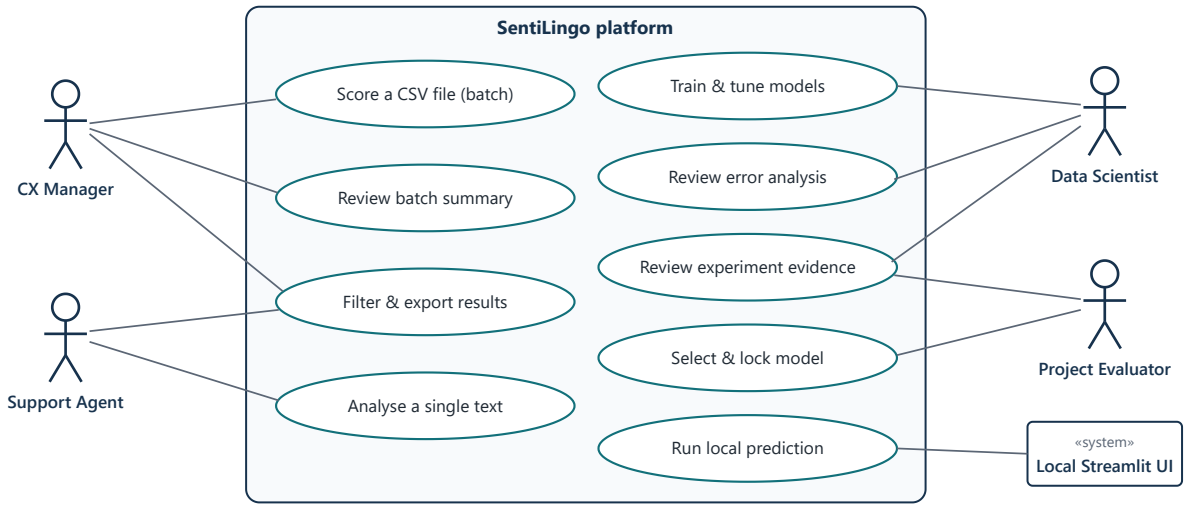


Figure 3 — Use-case diagram: business users consume predictions while technical users produce and review reproducible model evidence.

3.9 Traceability to the Brief

Brief item	Requirements	Deliverable
Project Planning, Stakeholder Analysis, Database Design and UI/UX Design	Part 1; 3.2; Parts 4–5	D1
Dataset, labels, licences and frozen splits	FR-01 – FR-04	D2
Text cleaning, normalisation and feature preparation	FR-05 – FR-07	D3
TF-IDF and focused classical baselines	FR-08, FR-09	D4
XLM-R and controlled optional comparison	FR-10, FR-11	D5
Validation selection, final evaluation and error analysis	FR-12, FR-13, FR-15	D6
Single-text and CSV Streamlit application	FR-16, FR-17	D7
Experiment evidence and reproduction instructions	FR-14; NFR-06	D8
README, model card and final technical documentation	FR-19	D9
Presentation PDF, video demonstration and rehearsal	Milestone MS5	D10
Optional: FastAPI, Docker or Azure deployment	FR-18	Stretch scope

PART 4

4. Database Design

A minimal experiment and model registry that preserves reproducibility without turning the five-week capstone into a database product.

4.1 Decision and Scope

Selected option. SentiLingo will use a small, vendor-neutral relational design to record dataset versions, experiment runs, metrics and generated artefacts. SQLite is the preferred implementation if persistence is added because it is local, zero-cost and easy to back up. The logical design remains portable to PostgreSQL or SQL Server.

The registry answers four practical questions: Which data produced a result? Which configuration was executed? How did it perform for each supported language? Which files belong to that run? It does not store user accounts, raw customer messages or prediction history.

Stored	Referenced, not copied	Deliberately excluded
Dataset version metadata and checksums	Source manifests and split files	User accounts, passwords, roles and sessions
Run configuration, seed, environment and timing	Model weights, tokenizers and reports	Raw customer text and personal data
Validation and locked-test metrics by scope	Confusion matrices and error-analysis files	Prediction history, analytics and application events

4.2 Conceptual Entity Relationship Model

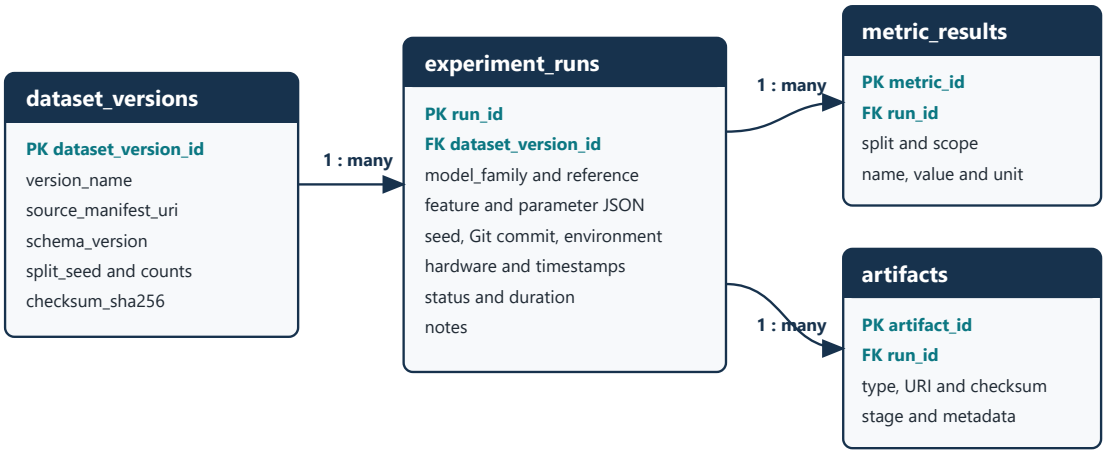


Figure 4 - One immutable dataset version can support many runs; each run owns its metrics and artefact references.

4.3 Logical Schema and Data Dictionary

dataset_versions

One row identifies the exact prepared data and frozen split policy used by experiments. Dataset files remain outside the database.

Field	Type	Rule	Meaning
`dataset_version_id`	TEXT	Primary key	Immutable identifier such as `ds_2026_11_08_v1`.
`version_name`	TEXT	Unique, required	Human-readable release name.

Field	Type	Rule	Meaning
`source_manifest_uri`	TEXT	Required	Path or URI to source, licence and provenance records.
`schema_version`	TEXT	Required	Version of the common dataset columns and validation rules.
`label_policy_version`	TEXT	Required	Version of the Positive, Neutral and Negative mapping policy.
`split_seed`	INTEGER	Required	Seed used for reproducible stratified splits.
`train_count`, `validation_count`, `test_count`	INTEGER	Non-negative	Audit counts after cleaning and deduplication.
`checksum_sha256`	TEXT	Unique, required	Detects an accidental change to the prepared dataset manifest.
`license_summary`, `created_at`	TEXT, TIMESTAMP	Required; UTC time	Usage limitations and creation time.

experiment_runs

One row represents one reproducible training or evaluation attempt. Configuration objects are stored as versioned JSON text to avoid adding tables for every algorithm-specific parameter.

Field	Type	Rule	Meaning
`run_id`	TEXT	Primary key	UUID or another collision-safe immutable identifier.
`dataset_version_id`	TEXT	Foreign key, required	Exact data version used by the run.
`run_name`, `model_family`, `model_reference`	TEXT	Name and family required	Readable name, algorithm family and base checkpoint or implementation.
`feature_config_json`, `hyperparameters_json`	TEXT	Valid versioned JSON	Vectorizer, preprocessing and training configuration.
`random_seed`, `git_commit_sha`, `environment_ref`	INTEGER, TEXT	Required when completed	Reproduction evidence for code and dependencies.
`hardware`, `started_at`, `completed_at`	TEXT, TIMESTAMP	UTC timestamps	Where and when the experiment ran.
`status`, `duration_seconds`, `notes`	TEXT, INTEGER, TEXT	Controlled status; non-negative duration	Lifecycle, elapsed time and concise observations.

metric_results

Field	Type	Rule	Meaning
`metric_id`, `run_id`	INTEGER, TEXT	Primary key; foreign key	Metric record and owning experiment.
`data_split`	TEXT	`validation` or `test`	Prevents test evidence from being confused with model-selection evidence.
`language_scope`	TEXT	Overall, English, MSA or Egyptian	Supports the approved language-balanced score.
`label_scope`	TEXT	All or one sentiment label	Allows overall and per-class evidence.
`metric_name`, `metric_value`, `unit`	TEXT, REAL, TEXT	Required; unique per scope	Macro-F1, accuracy, precision, recall, latency or another documented measure.
`is_primary`, `created_at`	BOOLEAN, TIMESTAMP	Required; UTC time	Marks the approved selection metric and record time.

artifacts

Field	Type	Rule	Meaning
`artifact_id`, `run_id`	TEXT	Primary key; foreign key	Artefact record and owning experiment.
`artifact_type`	TEXT	Controlled vocabulary	Model, tokenizer, confusion matrix, error analysis, sample predictions or report.
`artifact_uri`, `checksum_sha256`	TEXT	Required	Location and integrity check; the binary is not copied into SQLite.
`size_bytes`, `created_at`	INTEGER, TIMESTAMP	Non-negative; UTC time	Storage size and creation time.
`lifecycle_stage`	TEXT	Candidate, selected or retired	Records which model artefact is approved for the local demo.
`metadata_json`	TEXT	Valid versioned JSON	Optional framework, serialization and compatibility details.

4.4 Relationships and Integrity Rules

1. A dataset version may have many experiment runs; every run must reference exactly one dataset version.
2. A run may have many metric and artefact records. Orphan records are rejected by foreign keys.
3. Completed runs are append-only audit evidence. Corrections create a new run; identifiers and historical metrics are not overwritten.
4. The tuple of run, split, language scope, label scope and metric name is unique.
5. Counts, durations, sizes and metric values cannot be negative. Status and scope fields use controlled values.
6. Only a model artefact may be marked selected, and at most one model is selected for the current local-demo release.
7. Test metrics are written only after the configuration is locked. This is enforced by the experiment workflow and reviewed evidence.

4.5 Record Lifecycle

Stage	Database action	Evidence produced
Prepare data	Create one immutable dataset-version row after the audit and frozen split.	Manifest path, licences, schema and label-policy versions, counts and checksum.
Start a run	Create a run with status `running` and the full intended configuration.	Run ID shared by logs and output folders.
Finish or fail	Set completion time, duration and final status; retain failed runs when informative.	Reproducible history, including unsuccessful attempts.
Evaluate	Insert metrics by split, language and label; link confusion matrices and error analysis.	Comparable validation leaderboard and final locked-test evidence.
Select model	Mark the approved model artefact selected and retire any previous selection.	Traceable link from the demo model to its run and dataset.

4.6 Privacy, Security and Retention

- **Data minimisation:** store experiment metadata, not raw feedback or user identifiers.
- **Prediction logging:** disabled by default. The Streamlit application processes text in memory and does not add it to the registry.
- **Secrets:** never store tokens, passwords or cloud keys in the database or repository.
- **Access:** the local registry is writable only by the project team. Submission exports are read-only CSV or JSON files.
- **Retention:** keep completed-run metadata and checksums for reproducibility; delete unneeded model binaries separately without deleting their historical records.
- **Backup:** copy the SQLite file and exported registry at each milestone and before the presentation.

4.7 Physical Design and Portability

SQLite is the default physical store because the demonstration is local and concurrent writes are not expected. Foreign keys must be enabled. Timestamps use UTC ISO 8601 values, identifiers are text-based and JSON content carries an explicit schema version.

Physical choice	Decision
Indexes	Index runs by dataset and status, metrics by run and scope, and artefacts by run and type.
Generated files	The database and model artefacts are generated outputs and are not committed; schema documentation and export examples are committed.
Backup and export	Use a copied database file plus CSV or JSON exports so evaluators are not forced to install SQLite tools.
Migration path	Keep generic types and application-level JSON validation so PostgreSQL or SQL Server can replace SQLite later.

4.8 Alternatives and Consequences

Alternative	Why it was not selected	Consequence of skipping it
No operational database	Configuration files alone are simple but make cross-run comparison, integrity rules and audit queries harder.	The team accepts a small persistence task to gain database-design and reproducibility experience.
Full application database	Accounts, roles, authentication, prediction history and analytics would consume time needed for data quality and modelling.	The demo is single-team and local; it will not provide sign-in, personal history or multi-user concurrency.
Cloud-managed database	It adds cost, credentials, networking and availability risks without helping the required offline demonstration.	The first release does not demonstrate cloud database operations; portability is preserved in the logical design.
Store model binaries in the database	Large binary objects complicate backup, transfer and version control.	File storage must be organised and protected; the registry stores URIs, checksums and metadata instead.

4.9 Acceptance Checklist

- The ER model and all four tables are reviewed by the team.
- Primary keys, foreign keys, controlled values and unique metric scope are documented.
- No raw feedback, credentials or prediction history are included.
- One sample run can be traced to its dataset, metrics and artefacts.
- SQLite backup and CSV or JSON export can be demonstrated if persistence is implemented.

PART 5

5. UI/UX Design

A multi-page Streamlit dashboard that separates prediction, analysis and evidence while keeping sensitive feedback out of persistent storage.

5.1 Decision and Scope

Selected option. SentiLingo will use a multi-page local dashboard with sidebar navigation. This is the most portfolio-oriented option: business users can analyse feedback, while evaluators can inspect model evidence and experiment traceability.

Scope correction. A production administration area and persistent prediction history would contradict the approved minimal database and privacy boundary. The dashboard therefore keeps prediction results in the current session. Its history view contains model experiments only, never stored customer messages.

Core dashboard	Supported by the registry	Future product scope
Overview, text and CSV analysis	Dataset and model version references	User accounts and role administration
Current-session results and export	Experiment runs, metrics and artefact links	Persistent prediction history and saved dashboards
Model performance, help and limitations	Read-only selected-model traceability	Team workspaces, alerts and automated monitoring

5.2 Users and Design Goals

User	Primary goal	UX response
Support employee	Understand one message without ML knowledge.	One focused text input, labelled result, confidence explanation and visible limitation notice.
Customer-experience manager	Analyse a CSV and export row-level results.	Guided upload, column selection, validation preview, progress, filters and enriched CSV download.
ML engineer	Trace results and compare evidence.	Model-performance and experiment-registry pages with versions, splits, configurations and metrics.
Evaluator or recruiter	Understand scope, evidence and limitations quickly.	Overview, reproducible sample flow, honest evaluation separation and portfolio-ready navigation.

5.3 Information Architecture

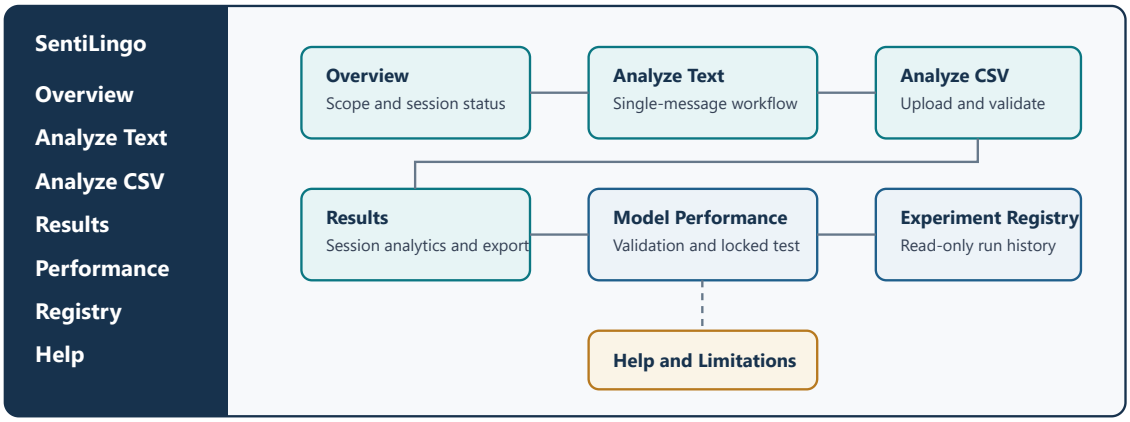


Figure 5 - Core prediction pages lead to current-session results; evidence pages expose model quality and reproducibility without storing customer text.

5.4 Primary User Flows

Flow	Steps	Completion evidence
Single text	Analyze Text -> enter or choose a labelled example -> validate -> analyse -> review label, confidence estimate, variety and limitation notice.	A clear result or an actionable inline error; no input is persisted.
CSV batch	Analyze CSV -> upload -> choose text column -> preview validation -> run -> Results -> filter -> download enriched CSV.	Every valid input row receives a result or row-level error; output preserves a row identifier.
Evaluator review	Overview -> Model Performance -> Experiment Registry -> Help and Limitations.	The selected artefact is traceable to its dataset, run, configuration and reported metrics.

5.5 Dashboard Wireframes

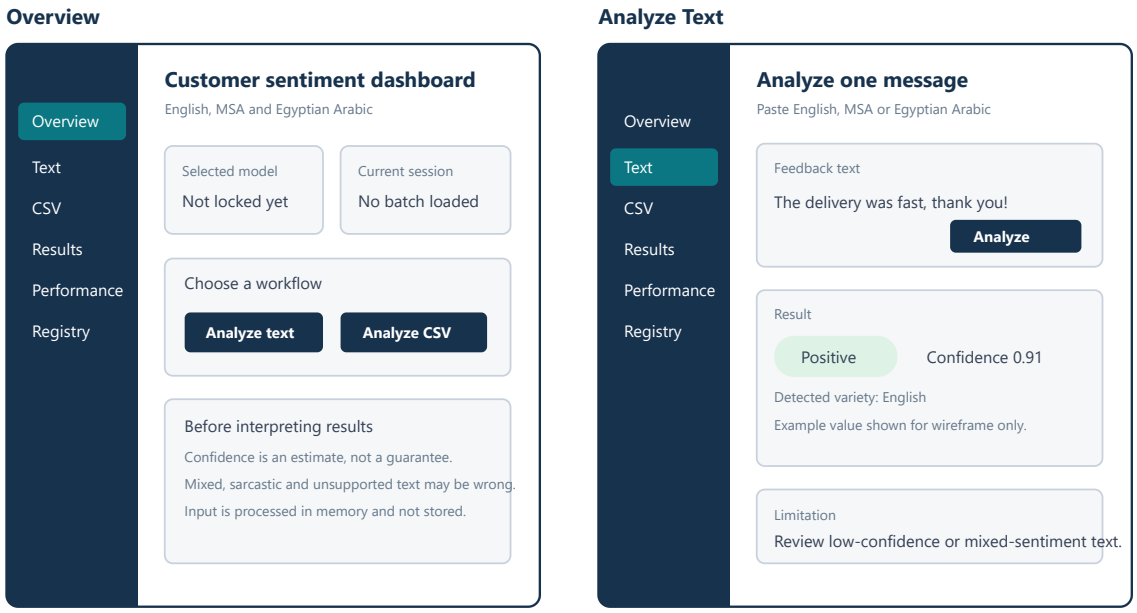


Figure 6 - Low-fidelity desktop wireframes. Scores and counts are illustrative placeholders until reproducible evaluation exists.



Figure 7 - Batch flow keeps validation, progress, analytics and export connected without creating persistent prediction history.

5.6 Screen Contracts

Page	Required content	Primary action	Important state
Overview	Scope, current model state, session state, privacy summary and workflow links.	Choose text or CSV analysis.	No model locked; no batch loaded.
Analyze Text	Examples, labelled text area, validation, result and limitation notice.	Analyse one message.	Empty, loading, success, unsupported or failed.
Analyze CSV	File input, text-column selector, validation summary, preview and progress.	Run a validated batch.	Wrong type, encoding failure, missing column or partial rows.
Results	Current-session filters, distribution, language breakdown, row table and export.	Download enriched CSV.	No active batch; partial success; export ready.
Model Performance	Model version, validation selection score, locked-test results, per-language metrics and limitations.	Inspect evidence.	Not evaluated; validation only; final test available.
Experiment Registry	Read-only runs, dataset version, commit, seed, hardware, metrics and artefact links.	Trace the selected model.	Registry unavailable; run incomplete; artefact missing.
Help	Labels, language scope, file contract, privacy, limitations and presentation fallback.	Understand correct use.	Always available offline.

5.7 Validation and Feedback States

- **Empty:** explain the next action with one realistic example; do not show fake analytics.
- **Loading:** disable duplicate submission and show the current operation without promising an unmeasured duration.
- **Success:** show the label, confidence estimate, language variety and model version together.
- **Partial success:** preserve valid CSV rows, identify invalid row IDs and allow export with error details.
- **Error:** state what failed, why when known and what the user can do next; never expose stack traces.
- **Unsupported or uncertain:** avoid silently forcing ambiguous, mixed-script or out-of-scope text into Neutral.

5.8 Accessibility, Arabic and Responsive Behaviour

- All inputs have visible labels, keyboard focus and error text associated with the affected control.
- Sentiment uses text and optional icons in addition to colour; contrast targets WCAG AA.
- Arabic message content uses right-to-left direction, while identifiers, numbers, file names and English controls remain left-to-right.
- The sidebar collapses at narrow widths; cards stack and result tables use a contained horizontal scroll only when necessary.
- Charts have text summaries and readable labels; essential values never require hover.
- Editable controls remain comfortable on touch devices and the workflow is testable at 360 px, tablet and desktop widths.

5.9 Visual and Data-Display Rules

Element	Rule
Navigation	Stable sidebar order, one active page and no duplicate page-level actions.
Sentiment	Always spell out Positive, Neutral or Negative; colour is secondary and consistent.
Confidence	Label it an estimate, show a numeric value only when the selected model supports it, and flag low-confidence results for review.

Element	Rule
Metrics	Keep validation and locked-test results visually separated; show per-language evidence before overall summaries.
Charts	Use directly labelled bars or tables, honest zero baselines where applicable and no decorative gauges.
Placeholders	Use “Not evaluated” or “Not available”; never invent scores, latency or model status.

5.10 Acceptance Checklist

- Every required page has one primary user goal and one clear primary action.
- Single-text, CSV and evaluator flows can be completed without hidden prerequisites.
- Prediction text is not stored and no page implies persistent customer history.
- Empty, loading, success, partial-success, error and unsupported states are defined.
- Arabic direction, keyboard access, contrast, responsive layout and offline operation are tested.
- Wireframes and final screens never present illustrative data as measured model performance.

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APPENDIX

Appendix A — Glossary

Term	Meaning in this document
Franco-Arabic (Arabizi)	Arabic written in Latin letters, with digits standing for Arabic sounds (e.g. 3 = ع, 7 = ح), as in `el service 7elw`.
Code-switching	Mixing two or more languages or scripts in one message.
MSA / Egyptian Arabic	Modern Standard Arabic, the formal written variety / the everyday Arabic dialect of Egypt.
TF-IDF	Term frequency–inverse document frequency: weights a word by how often it appears in a text, discounted by how common it is across all texts.
N-gram	A sequence of *n* consecutive words or characters, used as a feature.
BiLSTM	A bidirectional Long Short-Term Memory network: a recurrent model that reads text in both directions.
Transformer / fine-tuning	A neural architecture based on self-attention / further training a pretrained model on labelled task data.
Macro-F1	The average of the per-class F1 scores; every class counts equally, so minority classes are not ignored.
Confusion matrix	A table of predicted against true classes, showing which classes the model confuses.
MLflow	An open-source platform for tracking experiments and registering models.
CRISP-DM	A standard data-mining process: business understanding, data understanding, data preparation, modelling, evaluation, deployment.
MoSCoW	A prioritisation scheme: Must, Should, Could, Won't (this time).